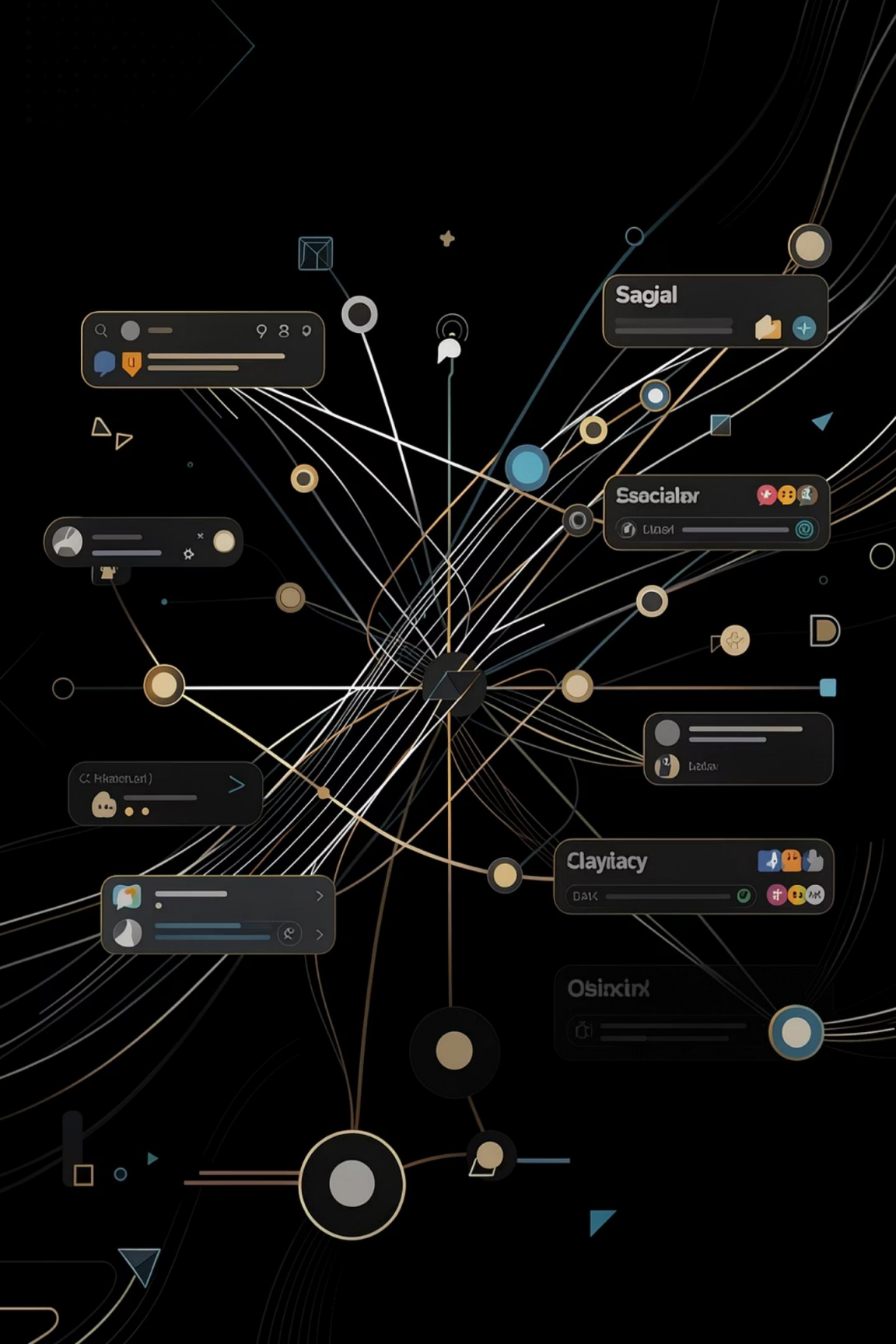




Social Media Clustering Pipeline

A step-by-step guide to extracting, processing, and clustering social media data — from API access to dimensionality reduction and unsupervised learning.



Step 1: Choose Your Platform

For this experiment, we use **Twitter (X)** as our social media data source. Twitter's developer API offers structured access to real-time and historical tweet data, making it ideal for NLP and clustering tasks.

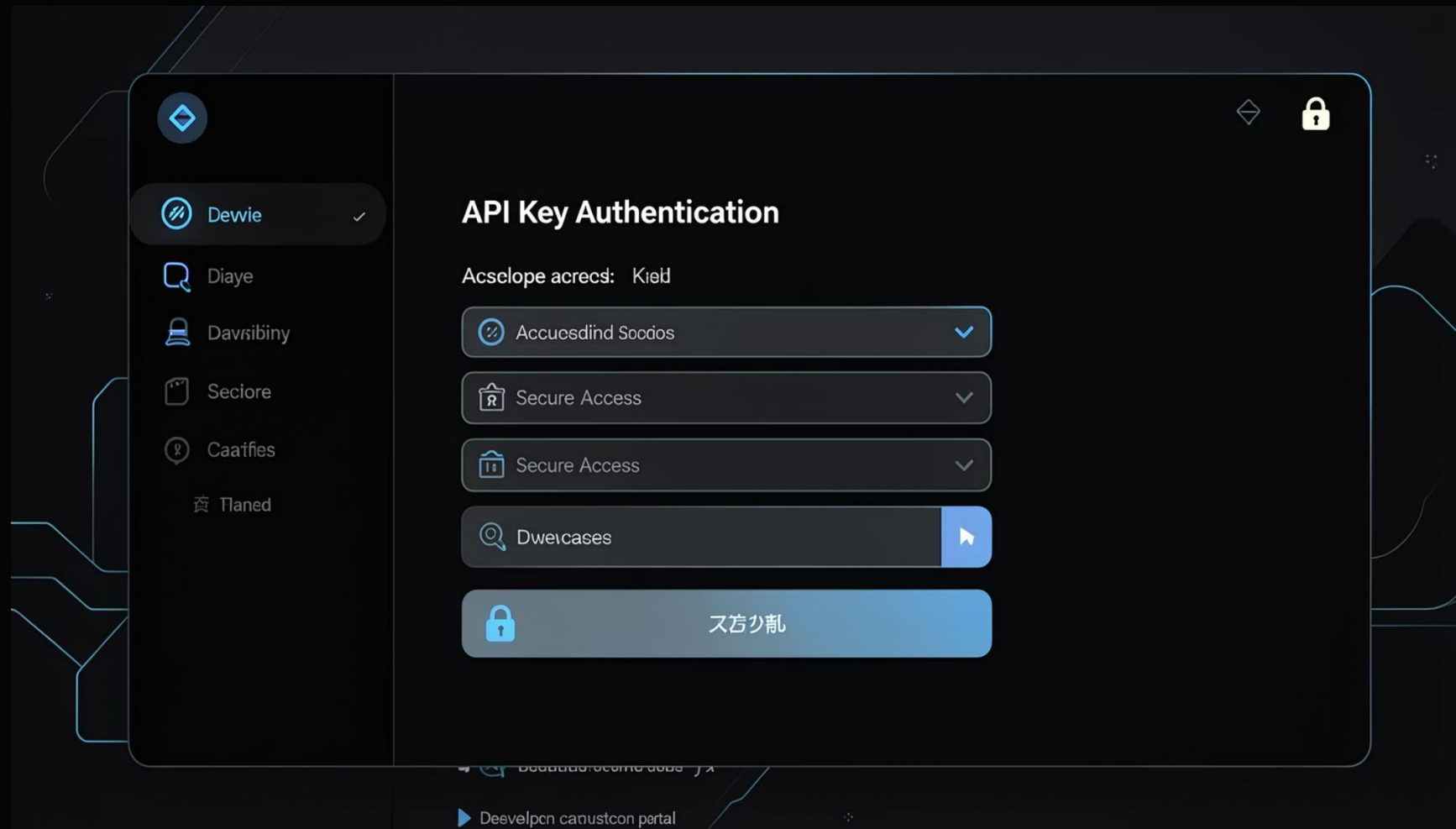
- i Other platforms (Reddit, YouTube) follow a similar approach — the pipeline generalises across sources.

Step 2: Obtain Your API Key

Register at the Twitter Developer Portal and create a project to receive your credentials:

- API Key & Secret
- Bearer Token (for read-only access)
- Access Token & Secret

⚠ Never hardcode credentials — store them in environment variables or a `.env` file.



Step 3: Extract the Data

Use the Twitter API (via `Tweepy` or `requests`) to collect a large corpus of tweets — targeting approximately **500,000 records** for statistically meaningful clustering results.

Query Parameters

Keywords, hashtags, language filters,
date ranges

Fields to Collect

Tweet text, user ID, timestamp,
retweet count, likes

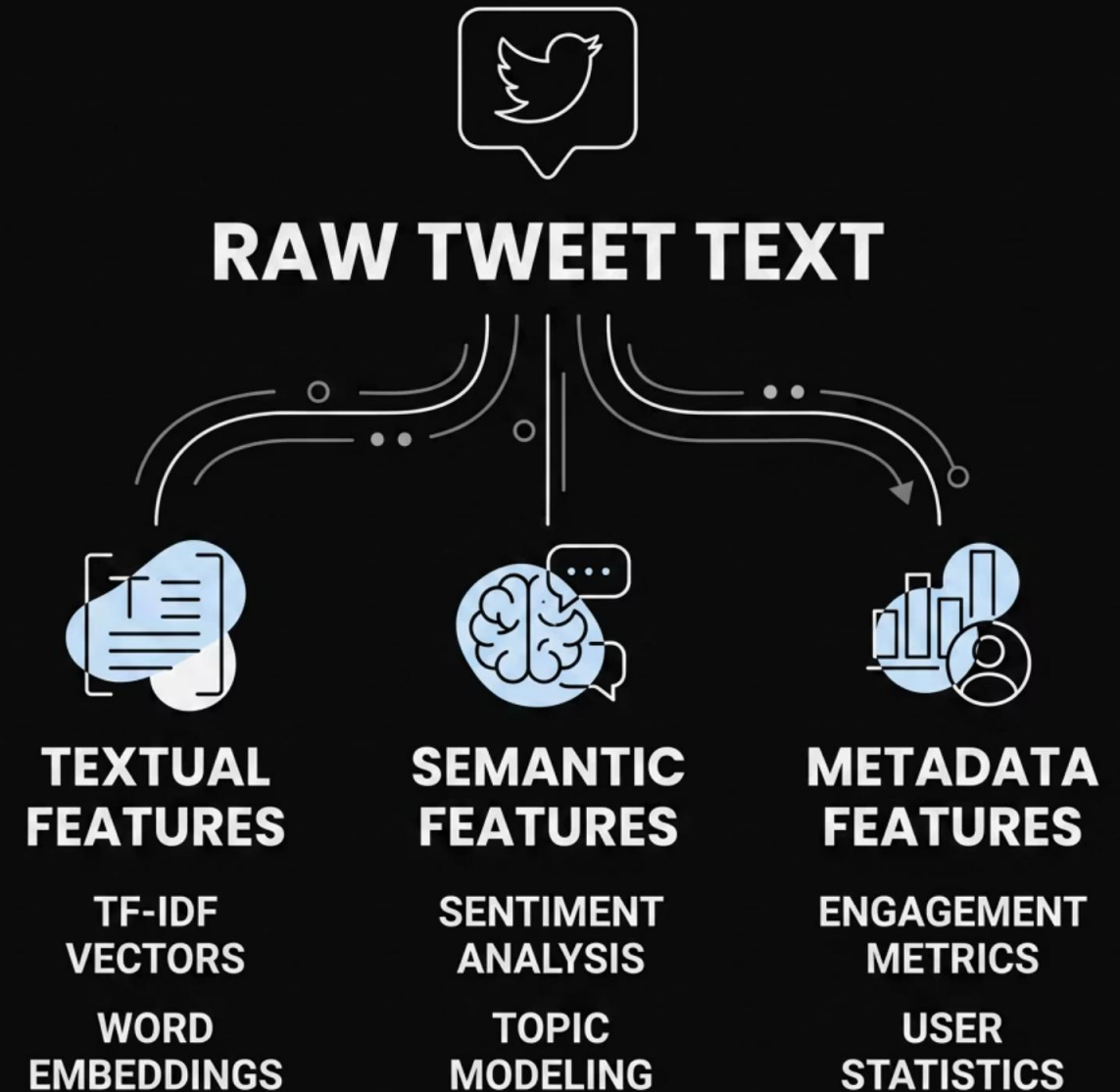
Storage

Save raw data as CSV or JSON for
reproducibility

Step 4: Feature Extraction

Transform raw tweet text into structured, machine-readable representations. Extract both **primary features** and **sub-features** to capture rich signal from the data.

- **Primary:** TF-IDF vectors, word embeddings (Word2Vec, GloVe)
- **Sub-features:** Hashtags, mentions, sentiment scores, emoji counts
- **Metadata:** Follower count, post frequency, engagement rate



Step 5: Data Visualisation

Before modelling, explore your data visually to understand distributions and patterns.



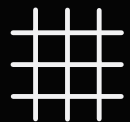
Frequency Plots

Top hashtags, most active users, tweet volume over time



Word Clouds

Visual summary of the most frequent terms in the corpus



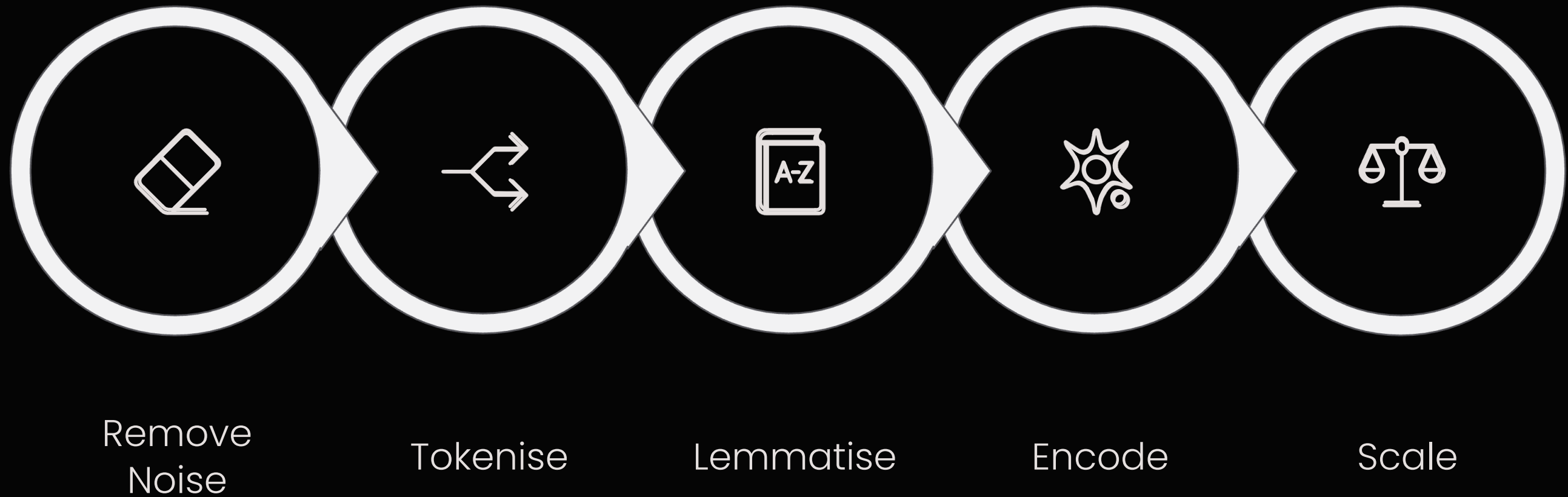
Correlation Heatmaps

Identify relationships between numerical features



Step 6: Preprocessing

Clean and normalise extracted features before applying dimensionality reduction or clustering algorithms.



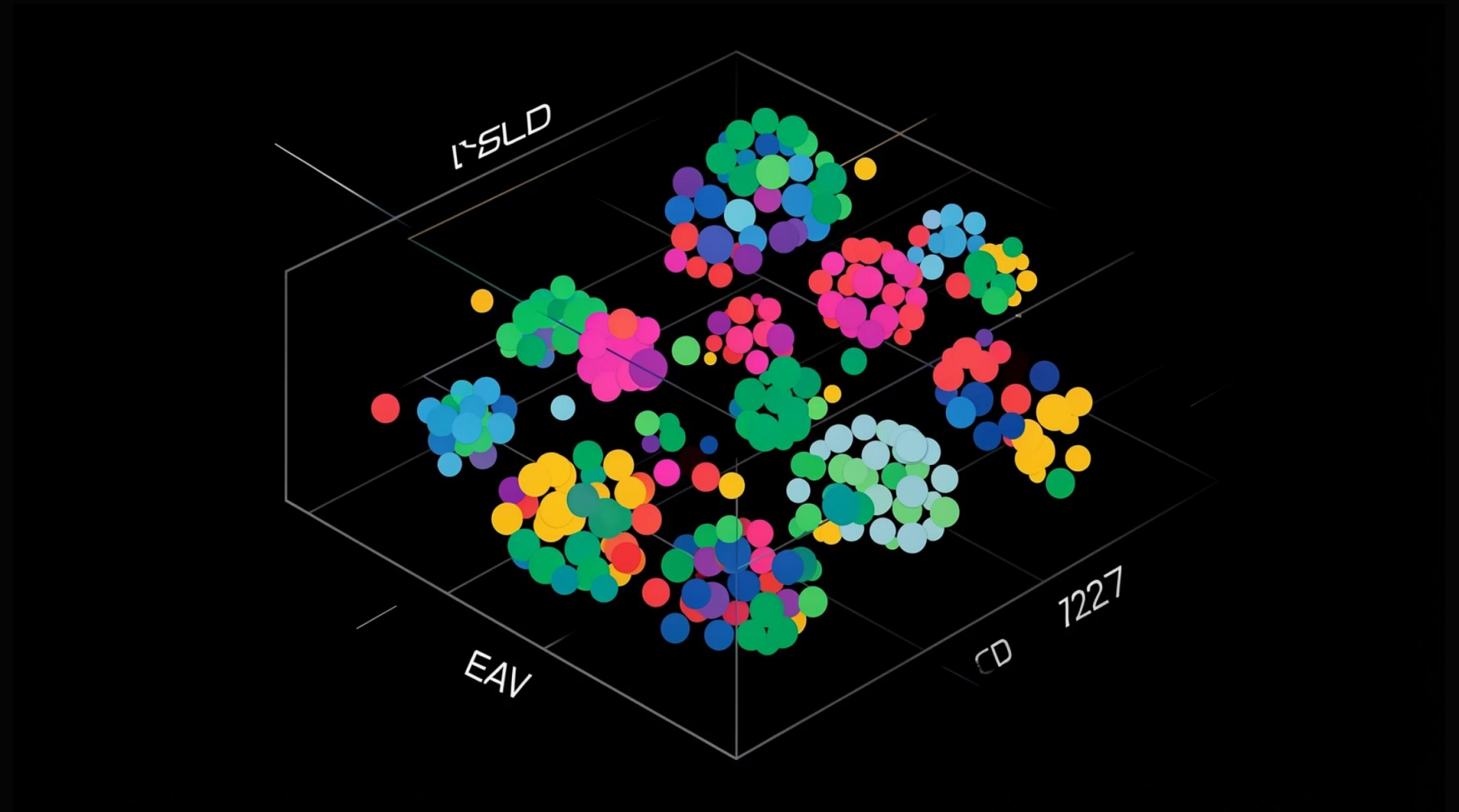
Consistent preprocessing ensures that your downstream models receive clean, comparable inputs — directly impacting clustering quality.

Step 7: Dimensionality Reduction

High-dimensional feature spaces are difficult to cluster effectively. Apply one or more reduction techniques to project data into a lower-dimensional space while preserving structure.

- PCA — linear reduction, fast, good for initial exploration
- SNE / t-SNE — non-linear, excellent for visualising cluster separation in 2D/3D

📌 t-SNE is preferred for visualisation;
PCA is preferred as a preprocessing
step before clustering.



Step 8: Clustering & Advanced Modelling

With reduced features in hand, apply unsupervised and deep learning techniques to discover meaningful groups in the data.

1

Clustering Algorithms

K-Means, DBSCAN, Agglomerative —
group tweets by topic or sentiment

2

BERT Embeddings

Use pre-trained BERT to generate
contextual sentence representations
for richer features

3

Transformer Fine-tuning

Fine-tune on domain-specific data to
improve topic coherence and cluster
quality

End-to-End Pipeline Summary

01

Platform & API

Select Twitter, obtain credentials

02

Extract & Explore

~500K tweets, visualise distributions

03

Feature Engineering

Extract, preprocess, and encode features

04

Reduce & Cluster

PCA / t-SNE → K-Means / BERT clustering

